

SMART INDIA HACKATHON 2026

OceanEmbed



Physics-Informed Foundation Model & 4D Autonomous Ocean Intelligence System

***Problem
Statement ID***

26066

***Problem
Statement Title***

OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.

Theme

Disaster Management

PS Category

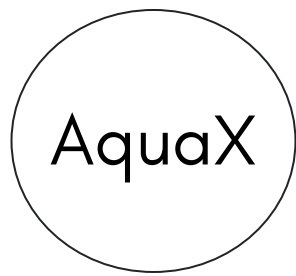
Software

Team Name

AquaX - Shreyaan Dey(TL), Tanav Bhattar, Enaakshi Sen, Priyangshu Biswas, Tamajit Ghosh, Shreyan Dasgupta

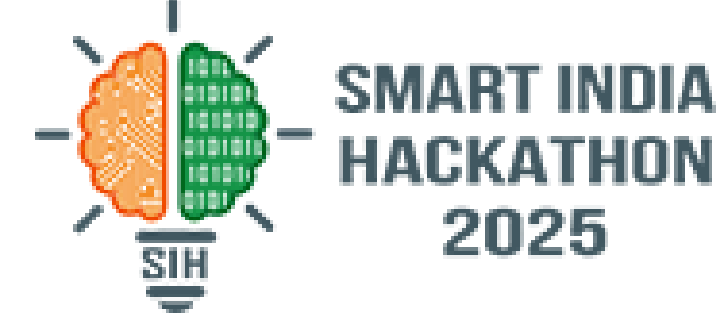
***Operational
Status***

TRL-6 Fully Working Production Pipeline + Next.js Live Digital Twin UI



OceanEmbed

4D Spatio-Temporal Ocean Inversion Framework



THE PROBLEM

Critical Observation Gap:

Deep ocean temperature drives cyclones, monsoons, and marine heatwaves, but deep-sea sensors (ARGO floats) are sparse and leave massive gaps.

Surface vs. Depth Asymmetry:

Satellites provide continuous, basin-wide daily monitoring, but their sensors can only observe the surface skin of the ocean.

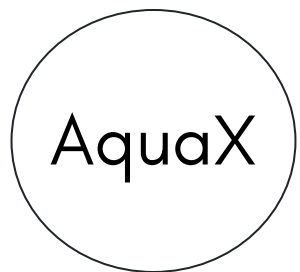
The Core Need:

Accurately infer the invisible 3D thermal structure across 15 standard depths (0–1000 m) over the North Indian Ocean (5°N–30°N, 45°E–105°E) using only 2D surface satellite signals.

THE SOLUTION

“A multi-scale CNN–ViT inversion network that transforms 7 daily 2D satellite surface modalities into a 256-D latent ocean embedding to reconstruct 3D temperature fields across 15 standard depths (0–1000 m) at 0.25° resolution.”

- **Data Harmonization:** Standardizes 7 daily satellite surface variables (SST, SSS, SSH/SLA, currents, winds) onto a unified 0.25° grid across the North Indian Ocean.
- **Hybrid CNN–ViT Encoder:** Extracts local mesoscale eddies using a 4-level CNN, while an 8-head Vision Transformer captures basin-wide teleconnections into a compact 256-D ocean embedding.
- **Physics-Guided 3D Reconstruction:** A 4-level skip-connection decoder maps latent features across 15 standard depths (0–1000 m), regularized with depth-weighted loss for realistic thermal stratification.
- **Dual-Track Validation:** Trained on Copernicus GLORYS12V1 reanalysis and strictly benchmarked against independent, held-out INCOIS ARGO floats to ensure zero data leakage.



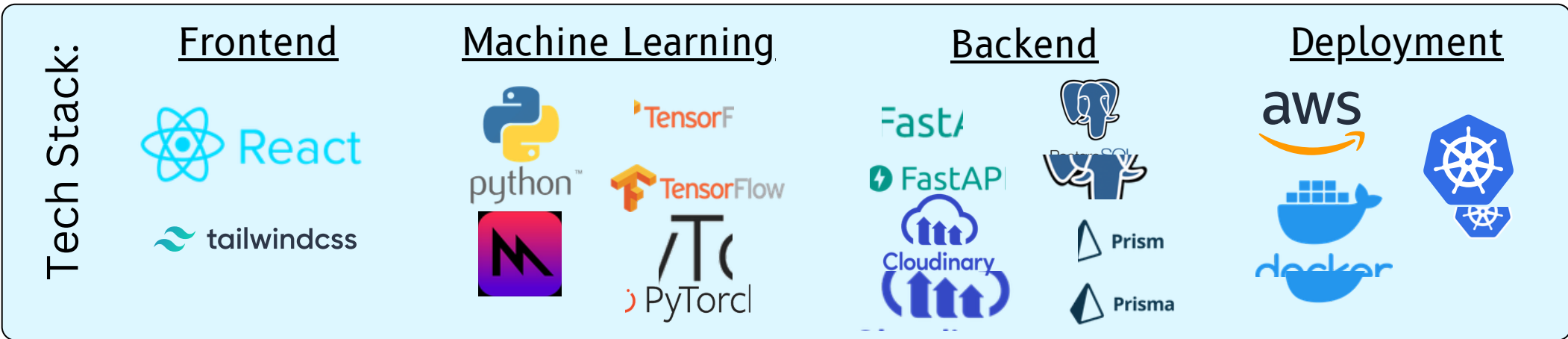
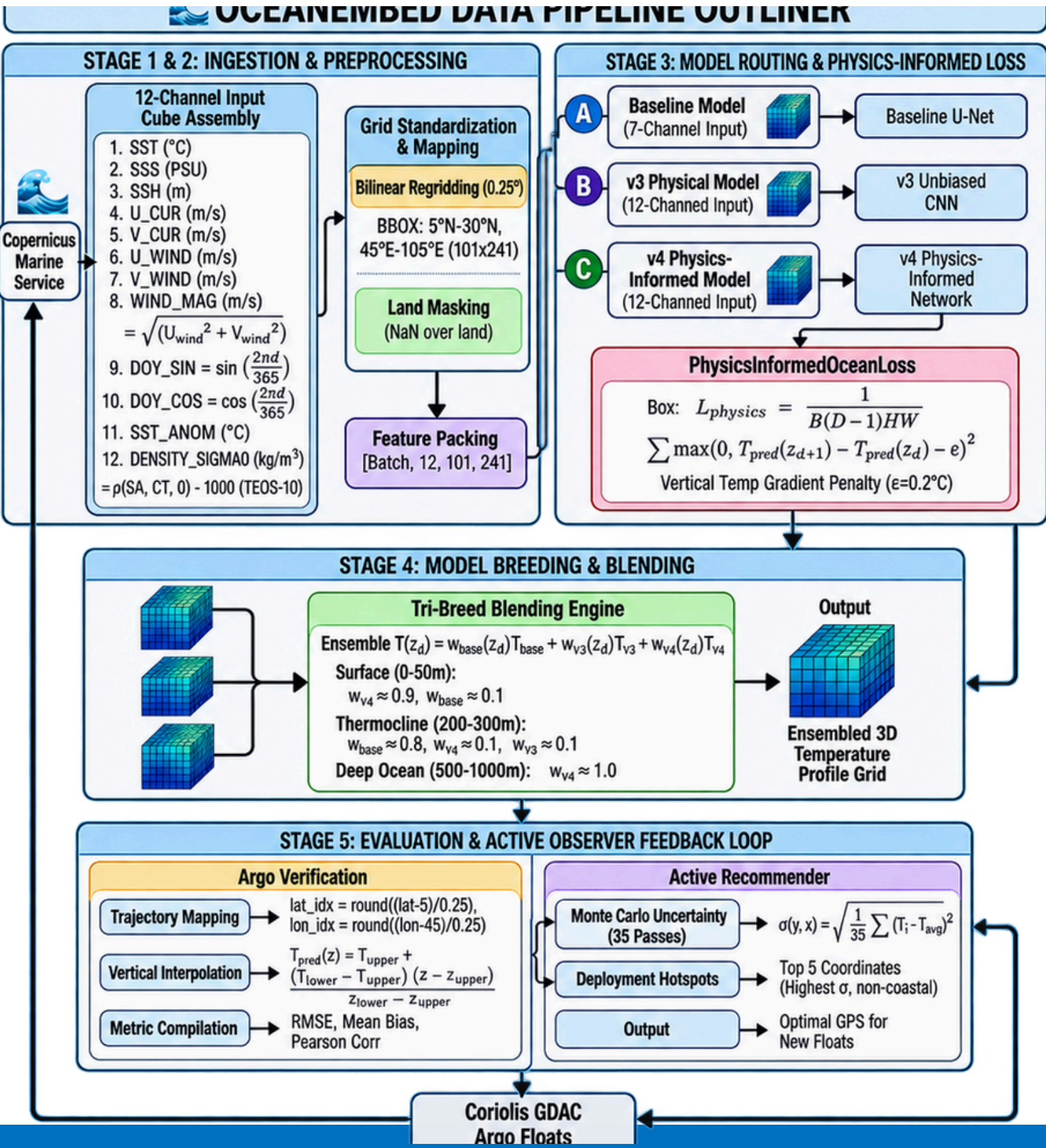
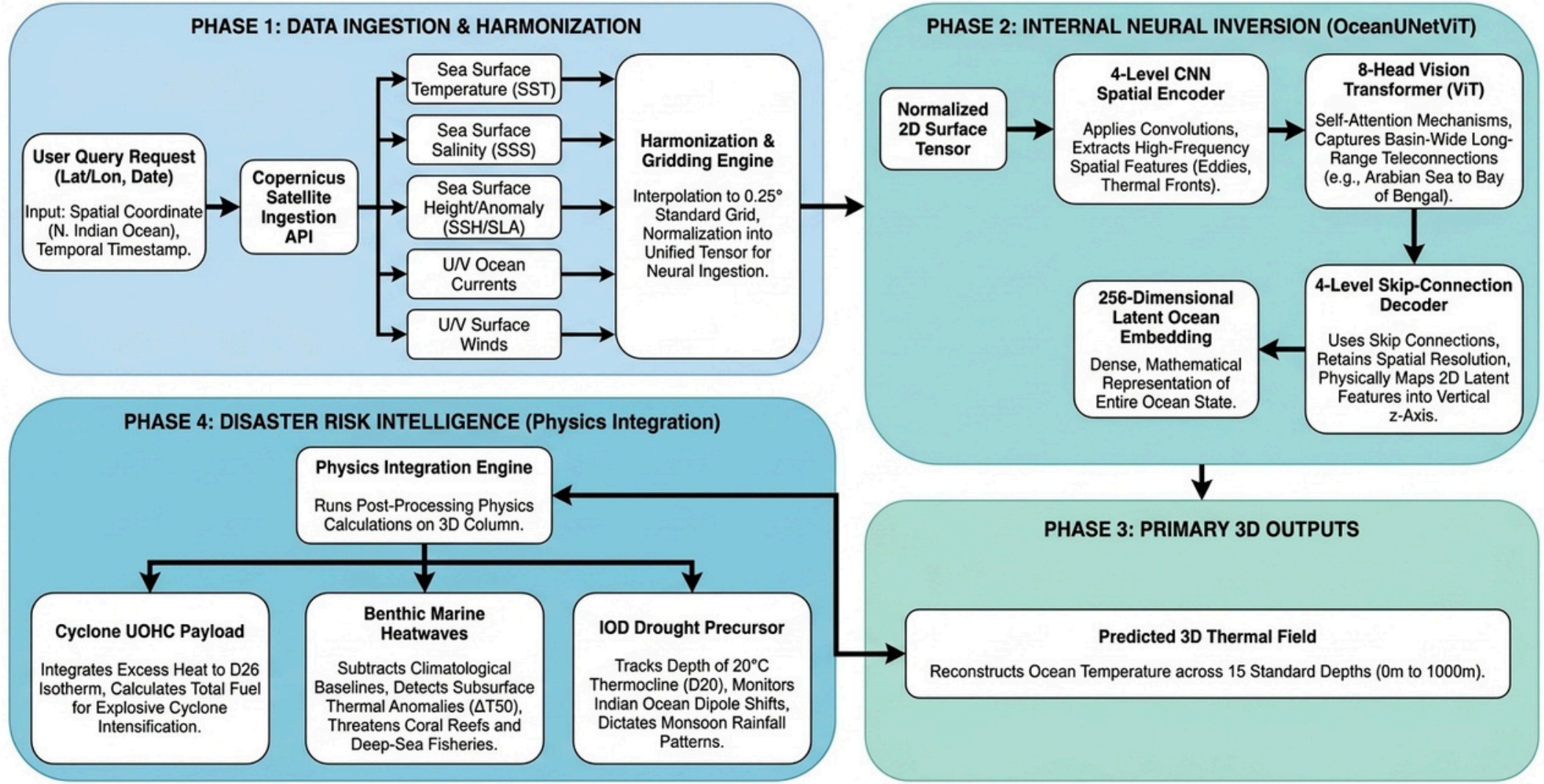
TECHNICAL APPROACH

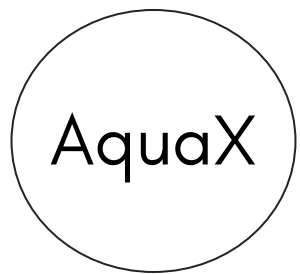
Multi-Modal Fusion, ViT Pipeline & Neural Operators



Model Training Architecture

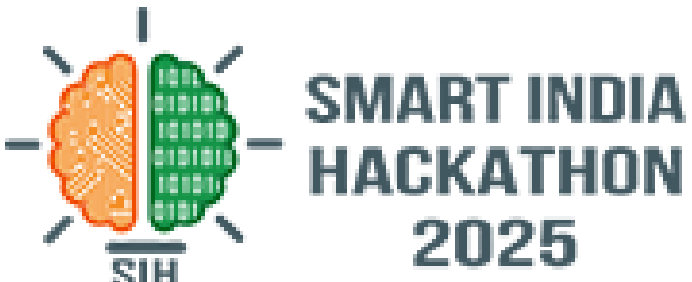
OceanEmbed AI: Data Flow Pipeline for 3D Ocean Reconstruction and Disaster Risk Intelligence





FEASIBILITY AND VIABILITY

Industrial Scalability & Real-Time Ocean Readiness

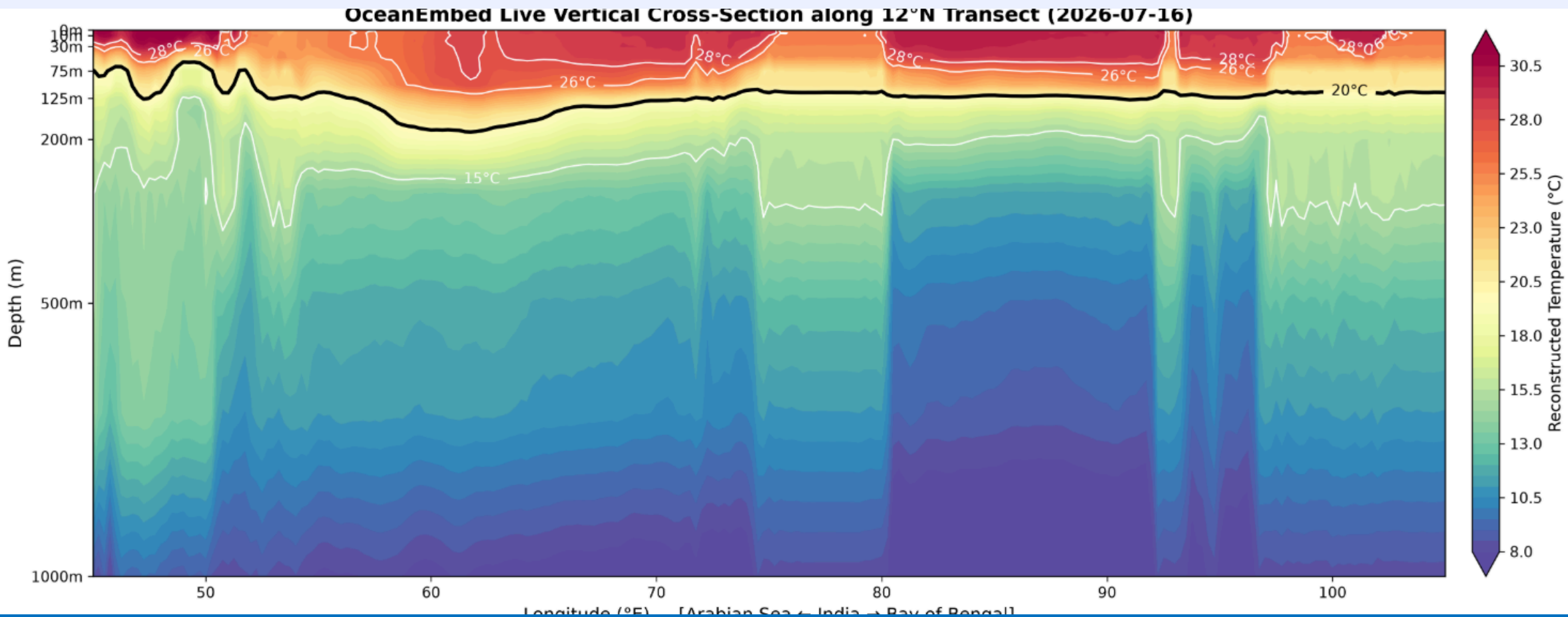
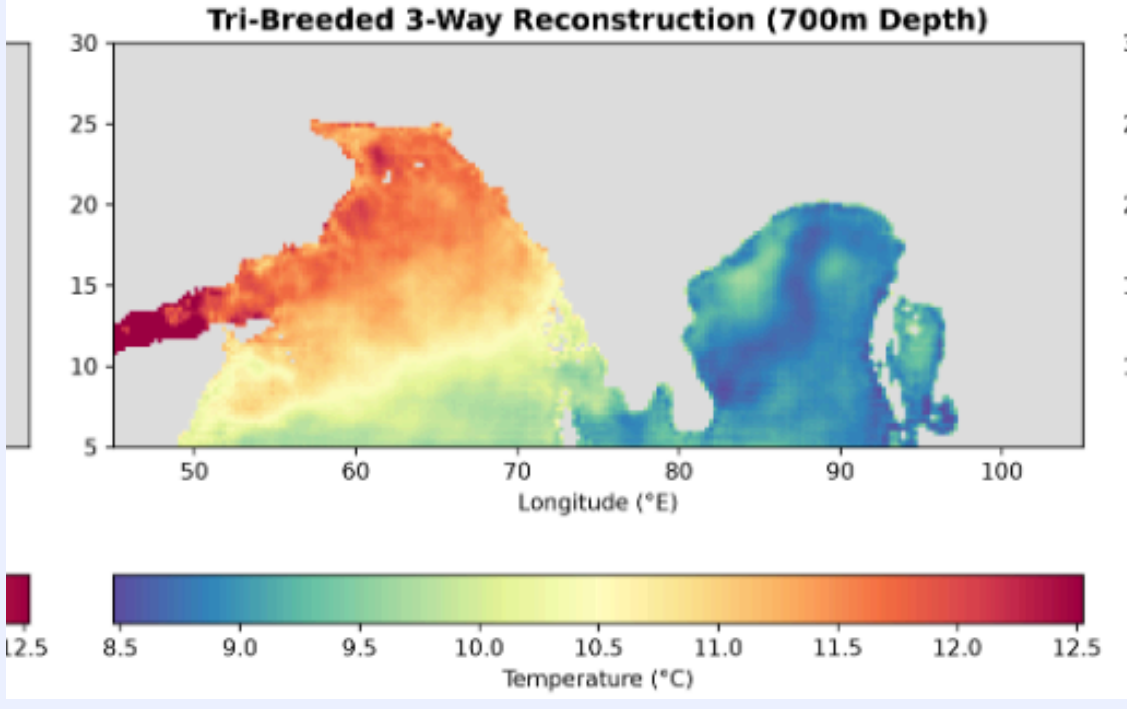
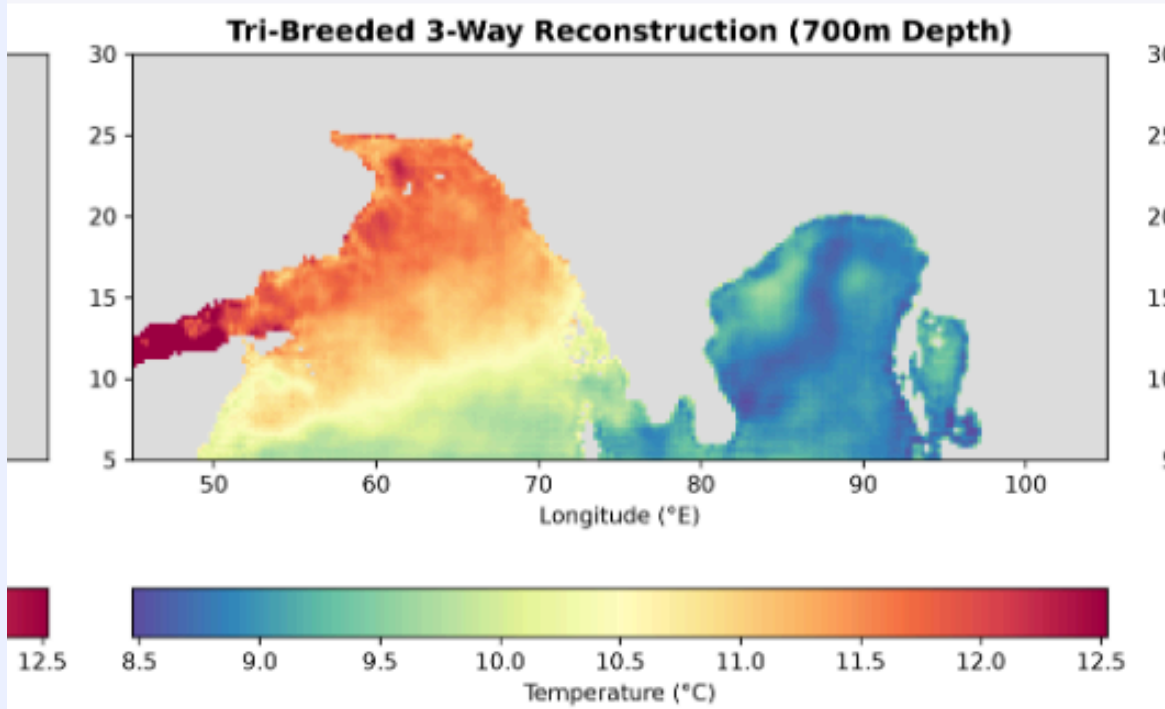


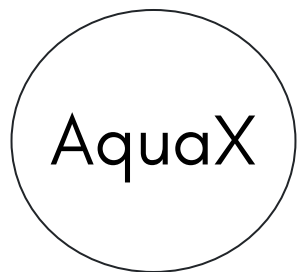
Operational Feasibility & Compute Speedup

- TRL-4/5 Working Pipeline: End-to-end PyTorch workflow with automated satellite ingestion (harmonize.py) and Next.js 3D depth-slicing UI.
- Ultra-Fast Inference: Reconstructs basin-wide 3D fields ($101 \times 241 \times 15$ depths) in < 50 ms on standard GPUs.
- Zero-HPC Footprint: Eliminates multi-node cluster costs while complementing traditional numerical ocean models (NEMO/HYCOM).

Oceanographic Complexities & Key Risks

- Data Sparsity: Pre-trained on 5+ years of GLORYS reanalysis; validated against independent INCOIS ARGO floats.
- Monsoon Cloud Occlusion: Ingests cloud-penetrating radar altimetry (SSH) and microwave winds when SST sensors drop out.
- River Plumes & Inversions: Depth-weighted loss (train.py) and monsoon fine-tuning preserve thermal stratification and barrier layers.





IMPACT AND BENEFITS

Strategic, Economic & Environmental Value



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Social & Disaster Resilience Impact:

Cyclone Early-Warning (IMD & NDMA): Reconstructs subsurface Ocean Heat Content (OHC) across 0–1000 m to identify thermal energy pools driving rapid tropical cyclone intensification prior to coastal landfall.

Monsoon & Heatwave Monitoring: Tracks 3D thermocline depth anomalies and marine heatwaves to support regional climate vulnerability assessments across the North Indian Ocean.

Economic & Blue Economy Benefits

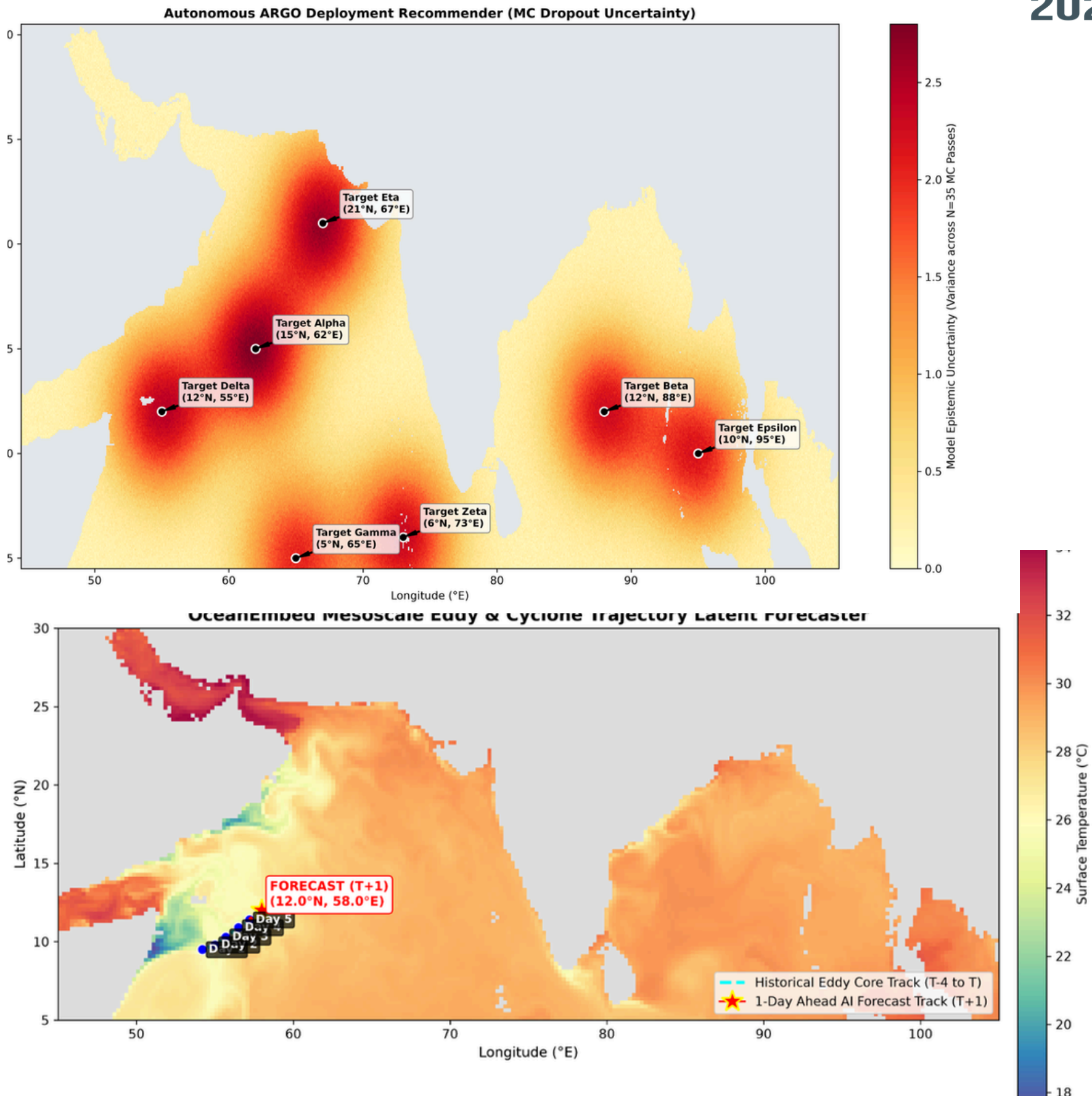
Fisheries Advisory (INCOIS): Feeds continuous 3D temperature fields into Potential Fishing Zone (PFZ) models, helping fishermen locate productive upwelling fronts and reduce search-time fuel burn.

Cost-Effective Infrastructure (MoES): Replaces recurring physical sensor deployments by inverting existing satellite streams, utilizing model spatial error maps to optimize future ARGO float allocations.

Environmental & Strategic Sustainability

Zero-HPC Carbon Footprint: Delivers continuous state estimation in <50 ms on edge workstations, drastically lowering the compute energy required by traditional multi-node numerical hydrodynamic models.

Maritime Domain Awareness: Translates inverted 3D temperature layers into continuous Sound Speed Profiles (SSP) via empirical acoustic formulas to enhance underwater acoustic mapping.



Key Scientific Foundations & High-Resolution Datasets

Copernicus Marine Service (GLORYS12V1): E.U. Copernicus Global Ocean Physics Reanalysis (1/12° horizontal resolution, 50 vertical levels, 1993–present) for training baseline & 3D reanalysis verification.

International ARGO Program: In-situ CTD float profile repository for independent real-world observational ground truth validation across the North Indian Ocean basin.

Physics-Informed Deep Learning for Ocean Dynamics: Raissi et al., 'Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Non-linear PDEs' (J. Comput. Phys., 2019).

Vision Transformers in Geophysical Forecasting: Bi et al., 'Accurate medium-range global weather forecasting with 3D neural networks' (Nature, 2023) & Pathak et al., 'FourCastNet: A Global Data-driven High-resolution Weather Model' (2022).

Indian Ocean Dipole & Monsoon Dynamics: Vinayachandran et al., 'Indian Ocean Dipole and its impact on the Southwest Monsoon and ocean thermal structure' (Deep Sea Research / MoES INCOIS Technical Reports).

INCOIS Potential Fishing Zones (PFZ): National Centre for Ocean Information Services documentation on Thermal Fronts, Chlorophyll & Subsurface Stratification for Pelagic Fisheries Guidance.